



MINING DEALER BEST PRACTICE

Hydraulic Wrench for CAT 330 Excavator Final Drive

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Hydraulic Wrench for CAT 330 Excavator Final Drive	1
1.0 Introduction.....	2
2.0 Best Practice Description	2
3.0 Implementation Steps.....	3
4.0 Benefits	12
5.0 Resources Required.....	13
6.0 Supporting Attachments / References	13
7.0 Related Best Practices	13
8.0 Acknowledgements	13

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

March 2025
4.03-250326-1

1.0 Introduction

To successfully assemble and disassemble the 330 Excavator side final drive, it is necessary to install the slide bearing in the housing and tighten the nut. However, the nut must be tightened to a torque of 2950 ± 295 lb-ft (4000 ± 400 N-m), which cannot be done with a hand tool because of the high torque value. Special equipment must be used to perform this operation.

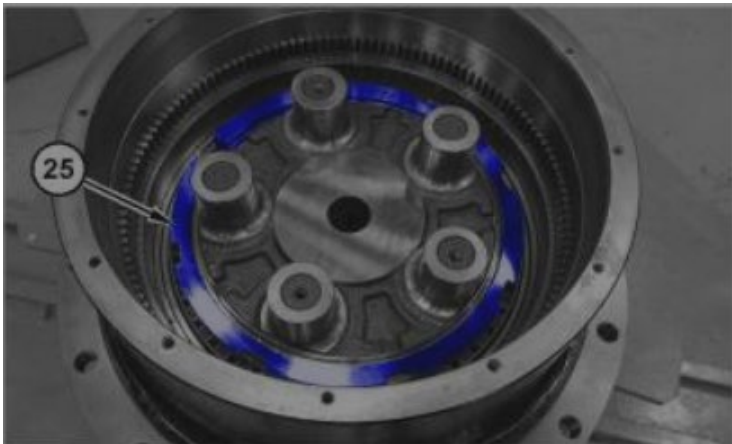


Fig. 1 Nut



Fig. 2 Stand for onboard final drive disassembly and assembly

Other dealers who assemble this onboard gearbox have to use a special bench to perform this operation. However, at our dealership, repairs on these bevel gears are quite rare, so we have developed an alternative assembly fixture that allows us to successfully perform this operation without the use of a special bench.

2.0 Best Practice Description

This Best Practice describes a tool that is designed to tighten the nut on an onboard final drive using a bench that is normally used for hydraulic cylinder assembly. Using it will avoid the need to purchase a specialized on-board final drive disassembly and assembly stand, as you can make an alternative stand yourself.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Hydraulic Wrench for CAT 330 Excavator Final Drive	DATE 3/26/2025	CHG NO 00	NUMBER 4.03-250326-1

To work with this wrench, you must use a special hydraulic cylinder assembly and disassembly stand. Follow the instructions below to complete the task:

1. Bolt the onboard gearbox to the bracket.
2. Lift the onboard final drive to stand height with slings and crane and secure it to the stand. Leave the slings taut for insurance.
3. Install the wrench in the hydraulic motor and lock it in place.
4. Start turning the hydraulic motor in the desired direction to tighten or loosen the nut.
5. Dismantle this construction from the stand.

When working with a hydraulic wrench, safety precautions and proper standards of procedure must be followed to avoid injury or damage. In addition, it is recommended that all necessary tools and accessories are checked, and that the equipment is visually inspected and tested before beginning work.

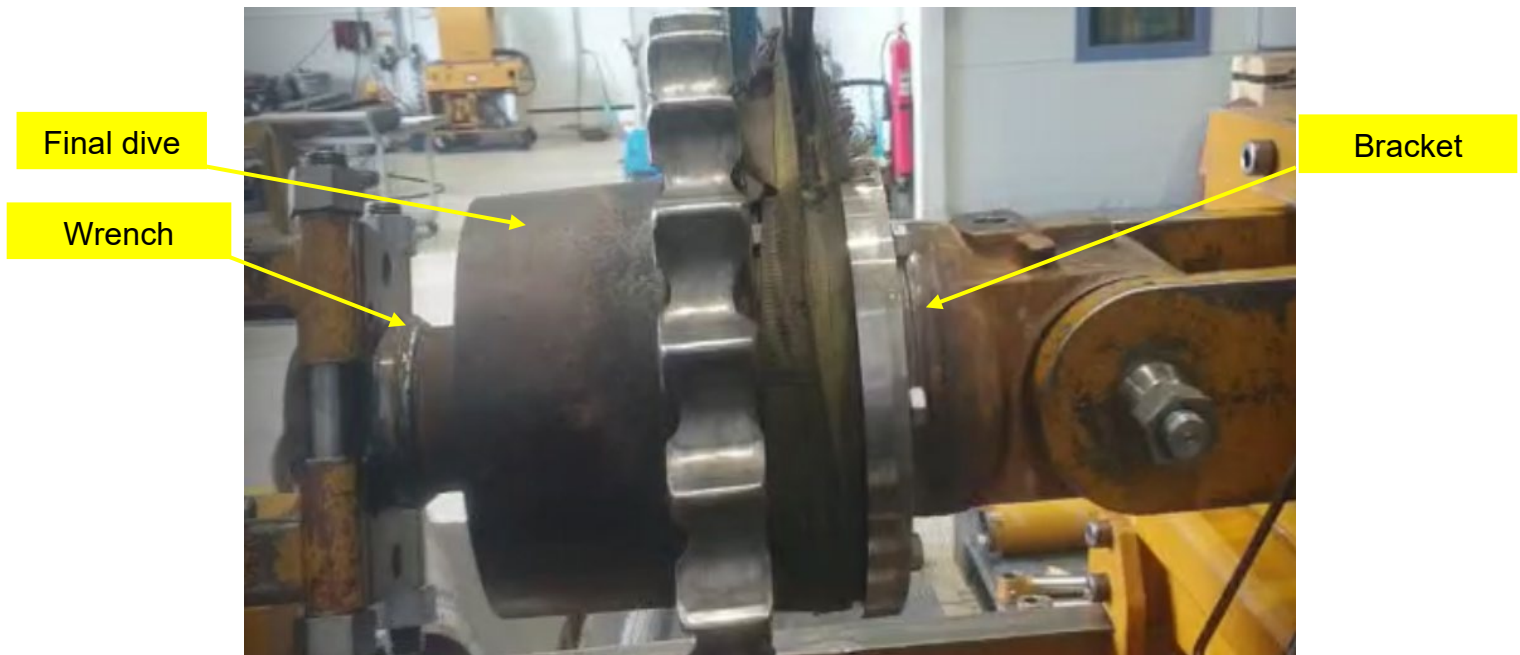


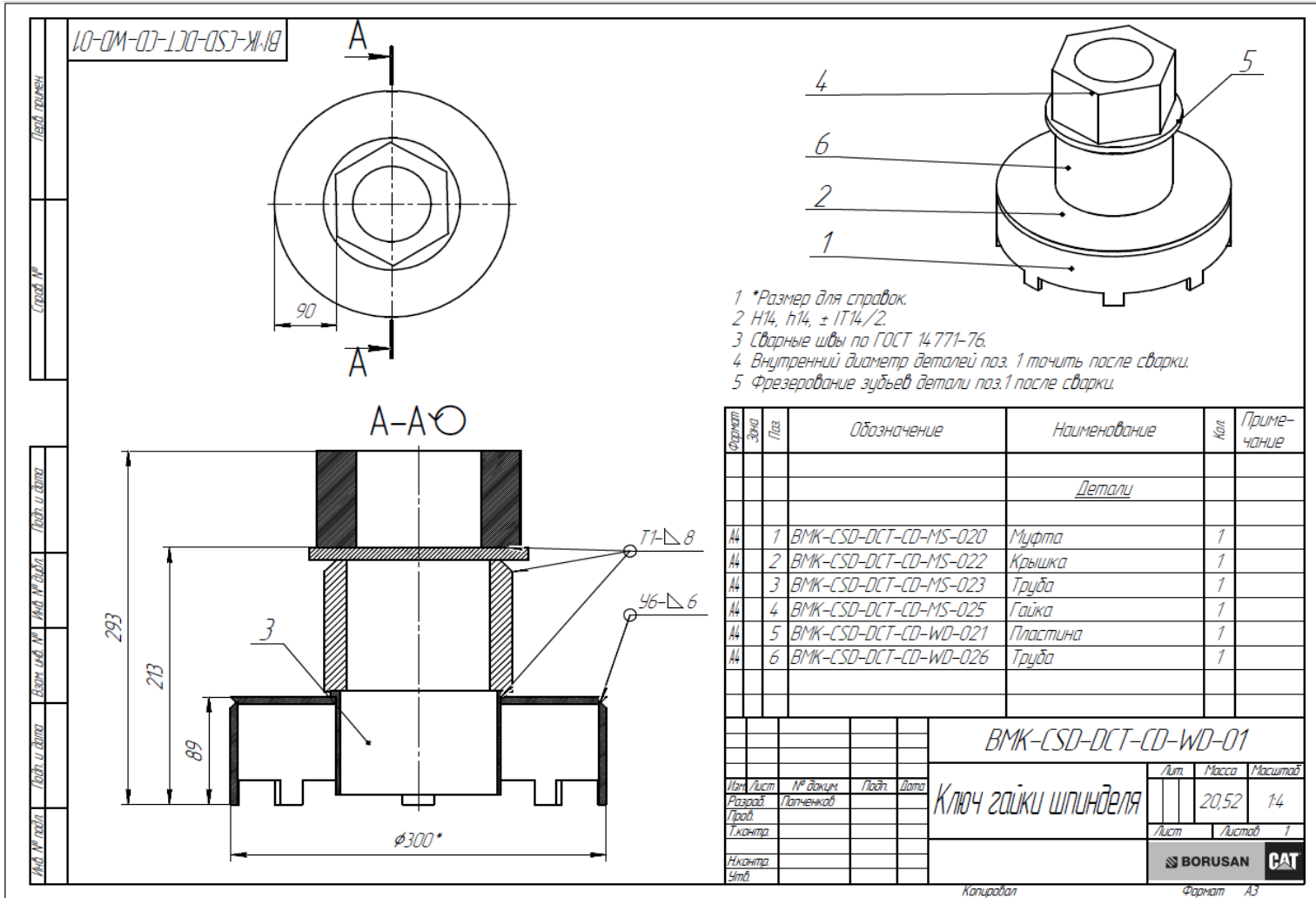
Fig. 3 tool for assembly, disassembly of the onboard final drive

3.0 Implementation Steps

This tool consists of two parts: in the form of a wrench clamped into a low-torque hydraulic motor mounted on the machine and a locking bracket mounted in the front headstock.

The wrench is made as follows:

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Hydraulic Wrench for CAT 330 Excavator Final Drive	DATE 3/26/2025	CHG NO 00	NUMBER 4.03-250326-1



Drawing 1 wrench

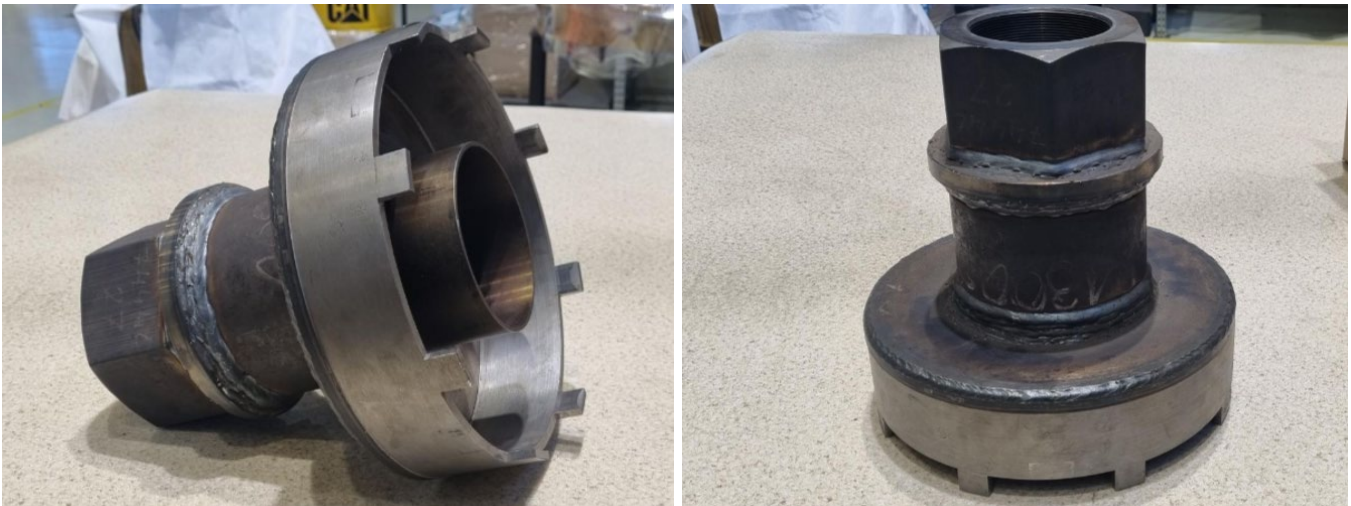
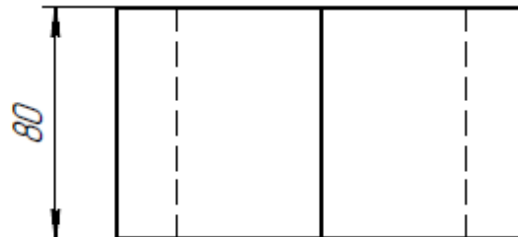


Fig. 4 wrench

Drawing 2 nut


$$1 \pm 1T14/2.$$

BMK-CSD-DCT-CD-MS-025

Гаўка

Сталь 10 ГОСТ 1050-2013

Лист	Масса	Масштаб
	6,04	1:2
Лист	Листов	1



BORUSAN

ТОО "Борусан-Астана"

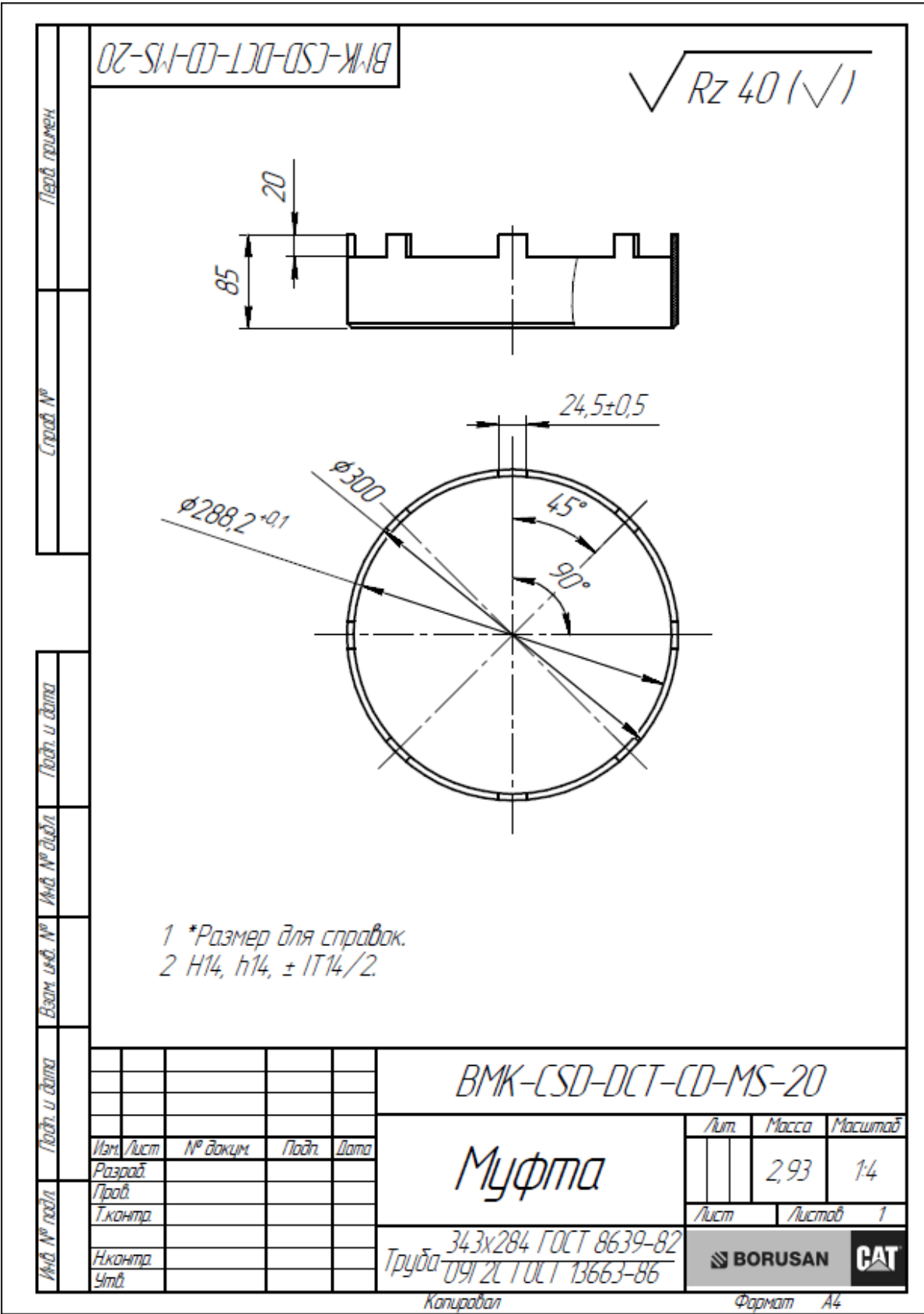


CAT

ТОО "Кат-Астана"

Копировать

Φορματίτη Α4



Drawing 3 coupling

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Hydraulic Wrench for CAT 330 Excavator Final Drive

DATE
3/26/2025

CHG
NO
00

NUMBER
4.03-250326-1

[illegible]

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

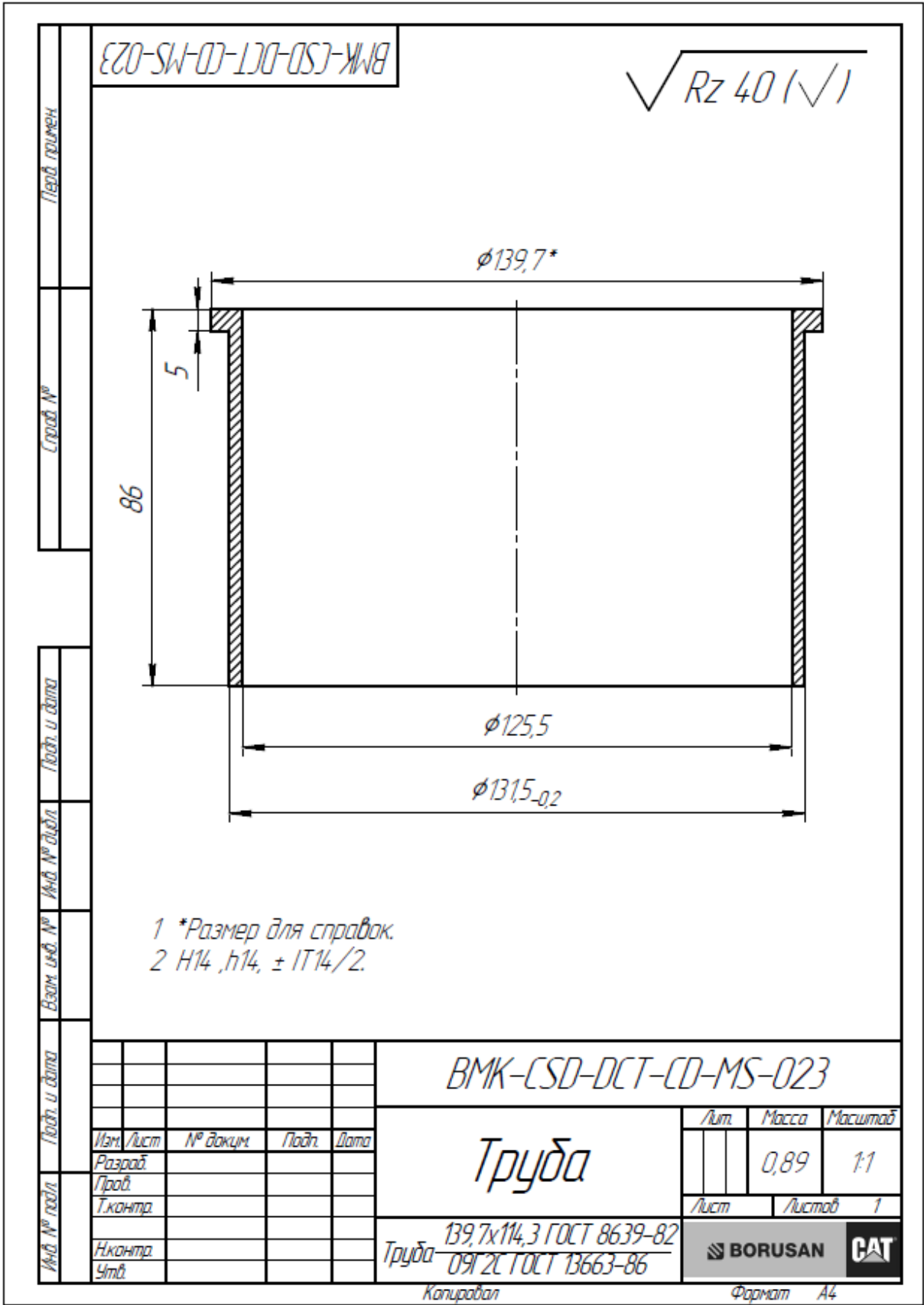
Hydraulic Wrench for CAT 330 Excavator Final Drive

DATE
3/26/2025

CHG
NO
00

NUMBER
4.03-250326-1

Drawing 5 pipe



Drawing 6 pipe

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

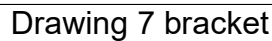
Hydraulic Wrench for CAT 330 Excavator Final Drive

DATE
3/26/2025

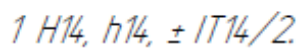
CHG
NO
00

NUMBER
4.03-250326-1

Bracket:



THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Hydraulic Wrench for CAT 330 Excavator Final Drive	DATE 3/26/2025	CHG NO 00	NUMBER 4.03-250326-1



BMK-CSD-DCT-CD-MS-002

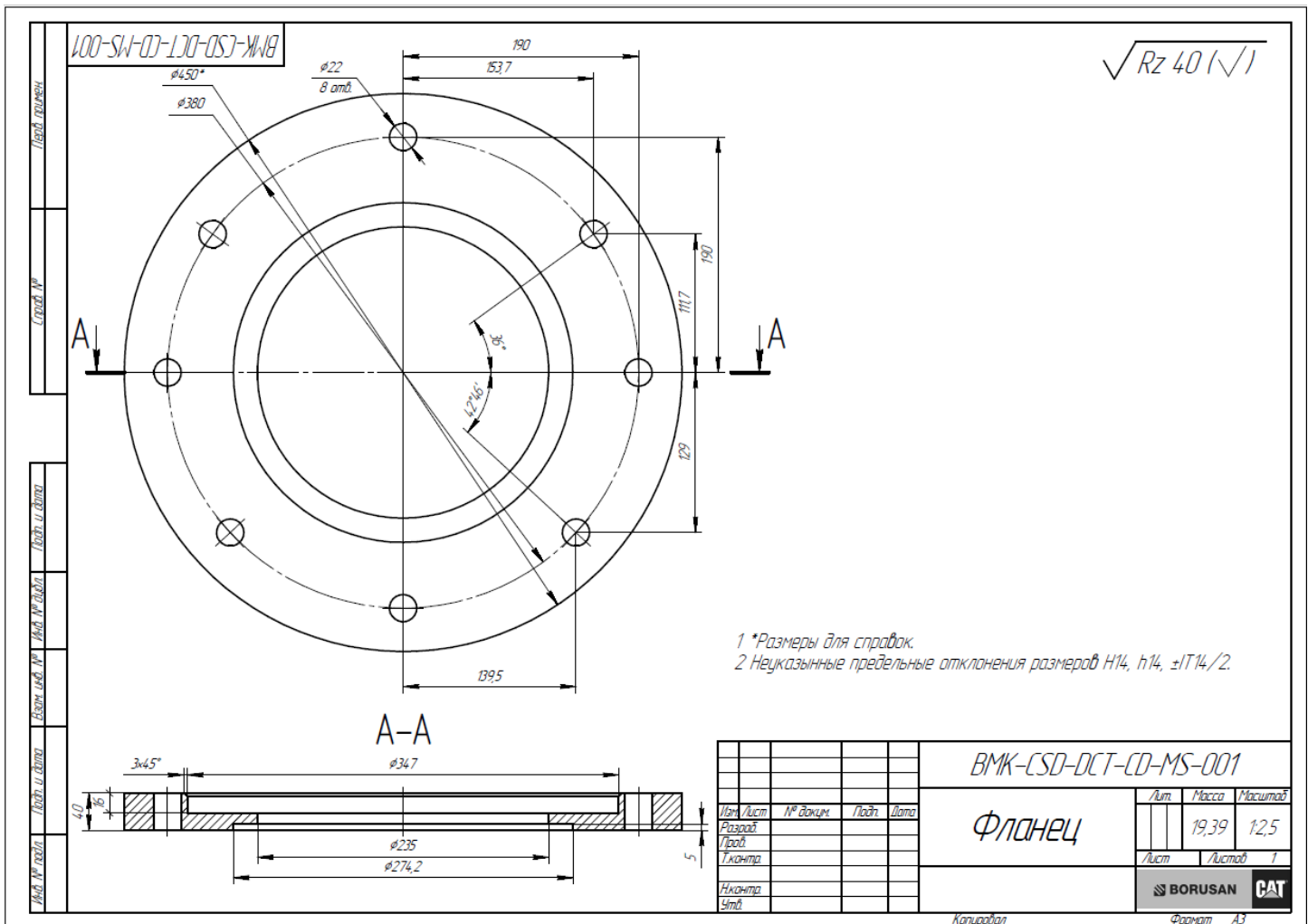
					Проушина(прототип)	Лист	Масса	Масштаб
Изм./Лист	№ докум.	Подп.	Дата				87,02	1:5
Разработ.								
Проб.								
Т.контр.						Лист	Листов	1
Н.контр.					Сталь 10 ГОСТ 1050-2013	BORUSAN		CAT
Утв.								

Καταγραφή

Формат A4

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

NUMBER
4.03-250326-1



Drawing 9 flange

Assemble the bracket according to the drawing.

The lug can be used from an old hydraulic cylinder with a suitable diameter.

4.0 Benefits

A tool consisting of a wrench clamped into a low-torque hydraulic motor mounted on the machine and a locking bracket has several advantages over buying a special stand:

Reduced costs. It is much cheaper to assemble a wrench set with a hydraulic motor and fixing bracket than to buy a special stand for assembly and disassembly of hydraulic cylinders separately which is available at many dealers. This allows you to save a significant amount of money on production equipment.

Space saving. Using this tool saves space in the shop, as it takes up much less space than a separate stand. This is especially important for small workshops where every square meter counts.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Hydraulic Wrench for CAT 330 Excavator Final Drive	DATE	CHG NO	NUMBER
	3/26/2025	00	4.03-250326-1

Versatility. This tool can be used with a variety of hydraulic motors and brackets, making it a versatile and flexible production tool.
In general, the use of this tool can significantly improve the efficiency and economy of production, reduce costs and simplify the work process.

5.0 Resources Required

- The machining shop and the time spent
- Hydraulic cylinder repair bench
- Consumption for metal
- The old nut from the hydraulic cylinder
- The old eyelet from the hydraulic cylinder

The manufacturing cost will be \$2,000.

6.0 Supporting Attachments / References

- see drawings

7.0 Related Best Practices

No

8.0 Acknowledgements

Writers:
Konobeitsev Kirill
Quality Engineer
Email: kkonobeitsev@borusan.com
Phone number 87768626561

Creators:
Bezuglov Algis
Head of the power transmission section
Email: abezuglov@borusan.com
Phone number: 87010829154

Papchenkov Maxim
Design Engineer
Email: mpapchenkov@borusan.com
Phone number: 87752738518

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Hydraulic Wrench for CAT 330 Excavator Final Drive	DATE 3/26/2025	CHG NO 00	NUMBER 4.03-250326-1